

Business Questions & Answers

Project Title: Understanding How E-Commerce Sales Work

Track: Data Analysis — Stage 7

Question 1: What are the top performing products by rating?

Rank	Product	Average Rating
1	Sync wire LTG to USB for fast charging	5.00
2	Amazon Basics Wireless Mouse 2.4 GHZ	5.00
3	REDTECH USB-C to lighting cable 3.3FT	5.00
4	Instant Pot Air Fryer Vortex	4.80
5	Swiffer Instant Electric Water Heater	4.80
6	Oratech Coffee frother Electric	4.80

Key Insight: The top 3 products all share a perfect rating of 5.00 — two are USB charging cables and one is an Amazon Basics wireless mouse. This shows that simple reliable accessories consistently receive the highest customer satisfaction. Products in the 4.80 range include kitchen and home appliances confirming that practical everyday products perform very well with customers.

Question 2: What is the relationship between actual price, discounted price and product rating?

Price Range	Average Rating	Average Actual Price	Average Discounted Price
Luxury (₹10,000+)	4.18	₹27,424.71	₹16,877.23
Mid-Range (₹501 – 2,000)	4.10	₹1,222.04	₹587.45
Budget (₹0 – 500)	4.07	₹362.83	₹195.16
Premium (₹2,001 – 10,000)	4.07	₹4,733.31	₹2,375.08

Key Insight: Across all price ranges the average rating remains consistently between 4.07 and 4.18 showing that price does not significantly impact customer satisfaction. Luxury products score the highest rating of 4.18 despite having the biggest gap between actual price (₹27,424.71) and discounted price (₹16,877.23) — a saving of ₹10,547. Budget products offer the smallest saving of ₹167.67 while still maintaining a strong rating of 4.07. This confirms that Amazon customers are satisfied across all price points and that heavy discounting across all categories adds significant perceived value.

Question 3: Which product categories have the highest average rating and highest number of reviews?

Category	Average Rating
Office Products	4.31
Toy & Games	4.30
Home improvement	4.25
Computer & Accessories	4.15
Electronics	4.08
Home & Kitchen	4.04
Health & Personal care	4.00
Musical Instrument	3.90
Car & Motorbike	3.80

Category by Total Review Count:

Category	Total Reviews
Electronics	15,778,848
Computer & Accessories	7,728,689
Home & Kitchen	2,990,077
Office Products	149,675
Musical Instruments	88,882
Toys & Games	15,867
Home improvement	8,566
Health & Personal Care	3,663
Car & Motorbike	1,118

Key Insight: There is an interesting contrast between rating and review count. Office Products has the highest average rating of 4.31 but only 149,675 total reviews. Electronics has the most reviews at 15,778,848 but a moderate rating of 4.08. This suggests that Electronics drives the most customer engagement and sales volume while Office Products delivers the best customer satisfaction. Car & Motorbike has both the lowest rating (3.80) and the fewest reviews (1,118) making it the weakest performing category overall.

Question 4: Which products are most popular based on user engagement?

Rank	Product	Total Rating Count
1	Amazon Basics High-speed HDMI cable	853,946
2	Amazon Basics Flexible Premium HDMI cable	853,945
3	boAt Bassheads 100 in Ear Wired Earphones	727,426
4	Redmi 9A sport (coral Green)	627,668
5	JBL C100SI Wired in Ear Headphones	577,766

Key Insight: Amazon Basics HDMI Cables dominate the top 2 positions with nearly identical engagement counts of 853,946 and 853,945 showing that Amazon's own brand products are extremely popular with customers. Audio products — boAt Earphones and JBL Headphones — appear in 3rd and 5th position confirming strong customer interest in affordable audio accessories. The Redmi 9A Sport smartphone in 4th place shows that budget smartphones also drive significant user engagement on the platform.

Question 5: KPIs and Additional Insights

Key Performance Indicators:

KPI	Value	Insight
Total Products	1,464	Large and diverse product catalogue
Average Rating	4.10	Strong Overall customer satisfaction
Average Discount	47.71%	Nearly half price savings on average
Average Price Saving	₹2,321	Significant Savings per product
Most Reviewed Category	Electronics	Highest Customer engagement

Additional Insight 1 — Discount Distribution:

Discount Range	Count	Percentage
High (41 – 60%)	514	35%
Very High (61%)	440	30%
Medium (21 – 40%)	339	23%
Low (0 – 20%)	171	12%

The majority of Amazon products — 65% — offer discounts above 40% showing that Amazon heavily uses discount pricing as a core strategy to attract customers. Only 12% of products have low discounts of 0-20% confirming that high discounts are the norm rather than the exception on the platform.

Additional Insight 2 — Price Range Distribution:

Price Range	Count	Insight
Mid-range (₹501 – 2,000)	523	Most common price range
Premium (₹2,001 – 10,000)	422	Second most common
Budget (₹0 – 500)	312	Third
Luxury (₹10,000+)	208	Least Common

Mid-Range products dominate the catalog at 523 products showing Amazon caters primarily to value conscious customers. Luxury products at only 208 confirm that Amazon's strength lies in affordable to mid-range pricing rather than premium luxury goods.